

Multi-Sensing Patch Combining Printed Electrodes And Paper-Based Millifluidics For Sweat Monitoring During Electrodermal Activity Monitoring

Sarah Tonello, Batoul Hosseinzadeh, Nicola Francesco Lopomo and Emilio Sardini

Abstract— Although Electrodermal Activity (EDA) is well-established technique for monitoring psychophysical conditions, its widespread adoption is limited by signal reliability challenges, primarily the difficulty in discriminating rapid phasic variations from slow tonic changes driven by personal features and workload. To bridge this gap, we propose a novel hybrid approach combining printed conductive electrodes with a millifluidic porous system to enable sweat absorption and monitoring over time. A multi-sensing patch integrating inkjet-printed electrodes with a laser-cut paper-based millifluidic network was developed. After optimizing geometry, ink, and substrate, the patch was tested in laboratory and industrial settings with real users, comparing results against commercial pre-gelled electrodes. Findings demonstrated that combining silver-based ink with PEDOT:PSS provides superior sensitivity and repeatability, while Whatman Grade 3 paper ensures optimal linearity. In vivo tests suggested performances comparable to the standard EDA commercial electrodes in terms of Pearson correlation coefficient between tonic (0.82 ± 0.09) and phasic (0.71 ± 0.13) components and of similar number of Skin Conductance Responses detected ($\Delta \text{SCR}/\text{min} < 3$). Overall, the proposed solution presents key innovative advantages in terms of complete customizability of the geometry, thus supporting the possibility to translate the approach to different wearables and personal protective equipment, representing a useful starting point for future physiological monitoring in a wide range of industrial workstations and operational conditions.

Index Terms— Electrodermal Activity, sweat, printed sensors, paper millifluidics, wearable devices, multi-sensing.

I. INTRODUCTION

OVER the last decade, wearable technology has experienced rapid diffusion, driven by the need for reliable health monitoring and a shift toward personalized, home-based healthcare [1], [2]. Despite advancements in living standards and health awareness have transformed wearables from basic activity trackers into comprehensive assessment tools, commercial devices mostly monitor basic parameters like heart rate and body temperature [3], [4]. However, since many diseases often involve complex and multifactorial biological changes, there is the need for

intensive, multiplexed data at a deeper level to enable better health monitoring through wearables. Recent advancements in smartwatches and electronic skins (e-skins) integrated with sweat sensors are addressing this gap by broadening the range of information non-invasively captured through the skin [5].

A field of peculiar interest is the evaluation of autonomic nervous system (ANS) function, specifically the activation of the sympathetic nervous system (SNS) [6]. Electrodermal activity (EDA) has emerged as a valuable biosignal for directly monitoring such functions across psychology, neurology, and cognitive science [7], [8]. The ANS regulates sweat secretion via eccrine sweat glands to maintain homeostasis in response to external or emotional stimuli [9]. Since sweat acts as a conductor, emotional excitation leads to a detectable rise in skin conductivity [10]. However, there is still a strong demand for accurate, wearable technologies capable of reliably quantifying EDA through these physiological reactions [11].

While sweating is essential for thermal balance, high sweat rates over long periods can cause critical losses of electrolytes [12]. For example, physical exercise without fluid replacement enhances the risk of dehydration and severe outcomes like heat stroke or reduced cognitive ability [13]. Considering this scenario, prompt and accurate monitoring of sweat both in terms of cumulative volume (μL) and of rate of loss ($\mu\text{L}/\text{min}$) could significantly optimize performance and prevent potentially dangerous health issues. Sweat analysis involves multiple steps, including collection, extraction, storage, and detection. Traditional methods for measuring sweat loss (both in terms of total volume in μL that of instantaneous rate in $\mu\text{L}/\text{min}$) include plastic bags, absorbent pads, or Macroduct systems [14], [15]. These techniques typically require complex laboratory procedures prone to analyte degradation [16]. To overcome these limitations, emerging technologies—including micro/millifluidics, hydrogels, epidermal systems or sealed chambers [17], [18], [19], [20], [21].—enable continuous, real-time monitoring via impedance, optical, or water vapor sensing [22], effectively minimizing evaporation during sampling. Currently, the gold-standard for EDA relies on Ag/AgCl electrodes with conductive gel, which face limitations in long-term durability. Dry electrodes are more practical for daily usage [23], [24], but developing flexible designs that maintain

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Final version of paper TIM-26-04647

reliable signal quality remains a challenge [25]. Printing techniques, particularly inkjet printing (IJP), are revolutionizing this field due to their cost-effectiveness, scalability, and ability to deposit functional materials in microscale patterns and over a wide range of flexible and stretchable materials [26]. Furthermore, the high geometric customizability of printing enables the rapid adaptation of electrodes to different body locations and integration into wearables like gloves or safety helmets [27] [28]. Accordingly, printing technologies can be exploited in order to obtain affordable, compact, and user-friendly solutions for extended wear.

A major limitation of wearable EDA systems is the difficulty in separating rapid phasic responses from slow tonic variations, especially in real world conditions where sweat production is influenced not only by sympathetic arousal but also by thermoregulation, physical workload, and local skin hydration. Accumulated sweat at the skin–electrode interface can artificially elevate the tonic skin conductance level (SCL), induce baseline drift, or generate pseudo responses that mimic phasic skin conductance responses (SCRs) despite the absence of sudomotor bursts [29]. Importantly, recent studies have shown that changes in skin hydration and sweat gland activity can alter skin conductance independently of sympathetic arousal, complicating the interpretation of EDA signals in real-world conditions where sweating is driven by both thermoregulation and emotional load [30].

In this context, simultaneous sweat volume and rate measurements provide the essential physiological grounding to distinguish true sympathetic activation from sweat accumulation artifacts in future signal processing pipelines. By quantifying the temporal evolution of sweat transport, sweat monitoring has the potential to support the contextualization of tonic drift, potentially accounting for hydration induced conductivity, and enhancing SCR detection under variable sweating conditions.

Integrating a millifluidic sweat sensor adjacent to dry EDA electrodes within the same patch offers a synergistic advantage, capturing local sweat dynamics at the same anatomical site to better understand their influence on the EDA signal. This dual sensing strategy aims to enhance interpretability, supporting a clearer decomposition of tonic and phasic components, and ultimately addresses one of the central limitations of wearable EDA systems—signal reliability in the presence of fluctuating sweat production.

In this study, we aimed to introduce a concept for a printed wearable patch fully integrated for combining preliminary EDA observations with information from continuous sweat quantification, focusing on estimating cumulative sweat volume (μL) over time, supported by an in vitro validation of the system's behavior under controlled flow conditions ($\mu\text{L}/\text{min}$). The sensor was fabricated using a cleanroom-free process, employing silver (Ag) electrodes in a coplanar two-electrode configuration. Sweat transport is achieved using paper-based millifluidic techniques, driving fluid through the channel via capillary-force gradients. This dual-function setup supports EDA tracking while continuously analyzing sweat flow. While preliminary in vivo tests are presented to demonstrate the device's technical feasibility, the primary focus of this study remains the in vitro characterization and the

optimization of the electrode fabrication process. The printed architecture proposed in this feasibility study facilitates straightforward geometric reconfiguration for different wearable formats and workplace settings, such as work gloves or helmet liners, offering a versatile design that could be adapted in future research for specific occupational environments.

II. PATCH DESIGN AND FABRICATION

A. Millifluidic design and fabrication

The paper-based millifluidic network was designed to enable capillary-driven sweat uptake and transport toward the sensing region, while minimizing premature evaporation and sample dispersion. The branched structure (*Figure 1*) was laser-cut on Whatman filter paper (Sigma-Aldrich) to exploit its capillary properties, enabling controlled and trackable fluid flow from a single inlet to the sensing region while reducing evaporation or sample loss before reaching the electrodes. The final layout consisted of a 4 mm-wide central channel, from which lateral branches extended symmetrically. Each branch (“tooth”) had a width of 3 mm, a length of 10 mm, and was separated from adjacent branches by a 2 mm gap.

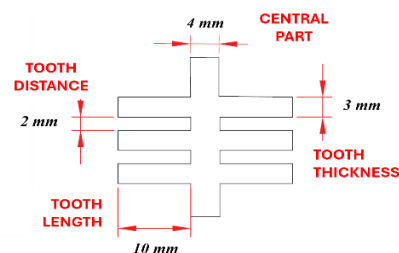


Figure 1: Design of paper-based millifluidics, presenting the following dimensions: central part: 4 mm; tooth thickness: 3 mm; tooth distance : 2 mm; tooth length: 10 mm.

The dimensions of this configuration were refined based on a comparative study of four configurations (short-thin, short-thick, long-thin, and long-thick teeth), reported in [31], which evaluated different geometrical layouts to balance flow rate, measurement sensitivity, and temporal stability. In summary, a comparative evaluation was performed using commercial electrodes, i.e., Ag/AgCl wet electrodes (Cardinal Health), combined with each of the configurations. Resistance between two electrodes was recorded while increasing volumes of phosphate-buffered saline (PBS, pH 7.4; Sigma-Aldrich) were introduced into the paper structures. Sensitivity, repeatability, and reproducibility of the resulting calibration curves were subsequently assessed. The characterization protocol was the same as the one detailed in Section 2.3 and Figure 3. Although 'short-thin' branches initially showed the highest sensitivity ($2.629 \mu\text{S}/\mu\text{L}$), they were rejected due to high variability (RSD of 46%). In contrast, the selected 'thick-teeth' geometry (3 mm) demonstrated lower variability (RSD < 15%) across the 0–100 μL range, with acceptable sensitivity ($2.201 \mu\text{S}/\mu\text{L}$), ensuring a more consistent contact interface with the electrodes. This stability can be reasonably attributed to the presumably more uniform contact interface provided by the thicker line designs, consistent with the observed reduction in RSD values. Based on these findings and considering that a lower RSD indicates a

Final version of paper TIM-26-04647

more consistent and accurate measurement system, thick-tooth patterns were selected as the most reliable configuration.

B. Inkjet printing fabrication of conductive electrodes

Inkjet printing was selected to fabricate the electrodes, since it offers high-resolution patterning, improved process flexibility, and prevents waste of precious material, being a mask-less drop-on-demand technique. With this goal, the inkjet printer Dimatix DMP-2850 (FUJIFILM Dimatix, Santa Clara, CA, USA) was used. The electrode design, substrate, and ink selection were based on the preliminary optimization reported in [31]. Square electrodes ($10 \text{ mm} \times 10 \text{ mm}$) spaced by 10 mm were chosen, since these dimensions previously demonstrated an optimal balance between sensitivity and noise during measurements (Figure 2). In addition to the primary pair of printed electrodes combined with the paper-based millifluidics channel for sweat tracking, an extra independent pair of dry electrodes was co-fabricated on the same coplanar patch substrate to allow simultaneous EDA monitoring at the same micro-environment site without fluidic crosstalk. Patterns were designed (Inkscape), exported as 24-bit BMP files, and then converted to monochrome BMP format, as required by the printing software. Polyimide film (Kapton, DuPont; $50 \text{ }\mu\text{m}$ thick) was selected as the primary substrate due to its excellent thermal, electrical, mechanical, and chemical properties, making it a widely adopted material in printed electronics. Thermoplastic stretchable polyurethane (TPU; Intexar, Micromax) was also evaluated as an alternative substrate, offering enhanced stretchability. Ink selection followed the main findings defined in [31]. The electrode optimization involved testing three configurations: bare silver (Ag), Ag/Carbon Nanotubes (CNTs), and Ag/PEDOT:PSS. While bare silver offered high reproducibility ($\text{RSD} < 10\%$), its sensitivity was insufficient. PEDOT:PSS was selected as the optimal coating because it achieved a superior sensitivity of $3.07 \text{ }\mu\text{S}/\mu\text{L}$, outperforming both CNT-based ($2.8 \text{ }\mu\text{S}/\mu\text{L}$) and standard pre-gelled wet electrodes ($2.4 \text{ }\mu\text{S}/\mu\text{L}$). As demonstrated by these results [31], PEDOT:PSS-coated electrodes provided the best overall performances, reducing differences between wet and dry electrodes and delivering superior sensitivity and repeatability in the $0\text{--}50 \text{ }\mu\text{L}$ range, while maintaining low contact impedance and reliable signal quality.

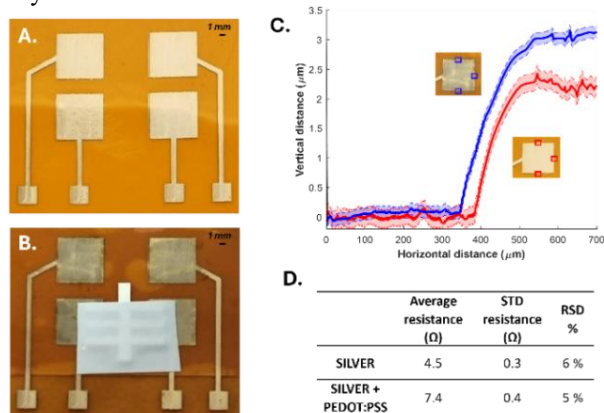


Figure 2: A. Printed Silver based electrodes and tracks; B. Printed patch combined with paper based millifluidics; C. Profile of printed electrodes with silver and silver-PEDOT:PSS coated; D. Resistance values of the printed electrodes

All printing and curing processes were conducted in a Therefore, silver and PEDOT:PSS inks were selected for fabrication. Silver nanoparticle ink (JS-A426, Novacentrix) and PEDOT:PSS ink (Orgacon™ IJ-1005, Sigma-Aldrich) were printed with 2 and 5 layers and with $10 \text{ }\mu\text{m}$ and $15 \text{ }\mu\text{m}$ drop spacings, respectively. controlled laboratory environment at $25 \pm 2^\circ\text{C}$ and $55 \pm 5\%$ relative humidity. For PEDOT:PSS, dimethyl sulfoxide (DMSO; Sigma-Aldrich) was added prior to cartridge filling to prevent clogging. The temperature of the printing plate was maintained at 60°C to ensure proper drying of each layer, avoid spreading due to multi-layer printing and ensure optimal adhesion between layers. After completing each ink deposition, thermal curing was performed according to manufacturer recommendations: 30 min at 250°C for silver, and 5 min at 110°C for PEDOT:PSS. Following printing and curing, electrodes were inspected optically, morphologically, and electrically to verify uniform ink deposition and ensure proper conductivity of the printed structures. Optical microscope evaluation reveals a printing tolerance of $\pm 100 \text{ }\mu\text{m}$ due to ink spreading on polyimide. Profilometer analysis (Fig. 2C) yielded average thickness of $2.2 \pm 0.2 \text{ }\mu\text{m}$ and of $1.2 \pm 0.1 \text{ }\mu\text{m}$ respectively for silver and PEDOT:PSS layers. Finally, connecting snaps required for measurements were bonded employing a silver conductive paste (Dycotec).

C. Laboratory-based characterization of printed electrodes

The laboratory-based characterization was organized in three main experimental phases: (i) selection of the most suitable Whatman paper grade, (ii) assessment under controlled flow conditions, and (iii) evaluation under bending stress to verify suitability for wearable applications.

The first set of experiments aimed to identify the optimal Whatman paper for the millifluidic pattern, as thickness and porosity can significantly affect sensor sensitivity and linear response. The optimized fluidic pattern, as described in Section 2.2, was fabricated using four paper grades: Grade 1 ($180 \text{ }\mu\text{m}$; $11 \text{ }\mu\text{m}$ pores), Grade 3 ($390 \text{ }\mu\text{m}$; $6 \text{ }\mu\text{m}$ pores), Grade 5 ($200 \text{ }\mu\text{m}$; $2.5 \text{ }\mu\text{m}$ pores), and Grade 6 ($180 \text{ }\mu\text{m}$; $3 \text{ }\mu\text{m}$ pores). Each paper grade was tested on three sensors. Comparative analysis was performed by injecting increasing volumes of PBS into the layered structure shown in Figure 3.D—comprising polyimide-based printed electrodes, the paper channel, and an insulating layer with a defined inlet—while continuously measuring resistance (Figure 3.C); this allowed us to correlate signal changes with respect to injected volume. Sensitivity, repeatability, and reproducibility of the calibration curves were evaluated. To explore compatibility with stretchable platforms, thermoplastic polyurethane (TPU; Intexar™ TE-11C, Micromax) was also tested as a substrate. TPU-printed electrodes were combined with the optimal paper grade and characterized using the same standard-addition protocol, comparing sensitivity and repeatability across five replicates against polyimide substrates. For flow evaluation, a peristaltic pump (Lambda, Merck) delivered PBS to the inlet at 10, 15, 25, and $30 \text{ }\mu\text{L}/\text{min}$ (Fig. 3B) while resistance was recorded for 15 min (Fig. 3C). To assess conformability for wearable integration, the sensors were tested under curvature using cylinders of 12, 16, 20, and 25 cm diameters via the same standard-addition protocol.

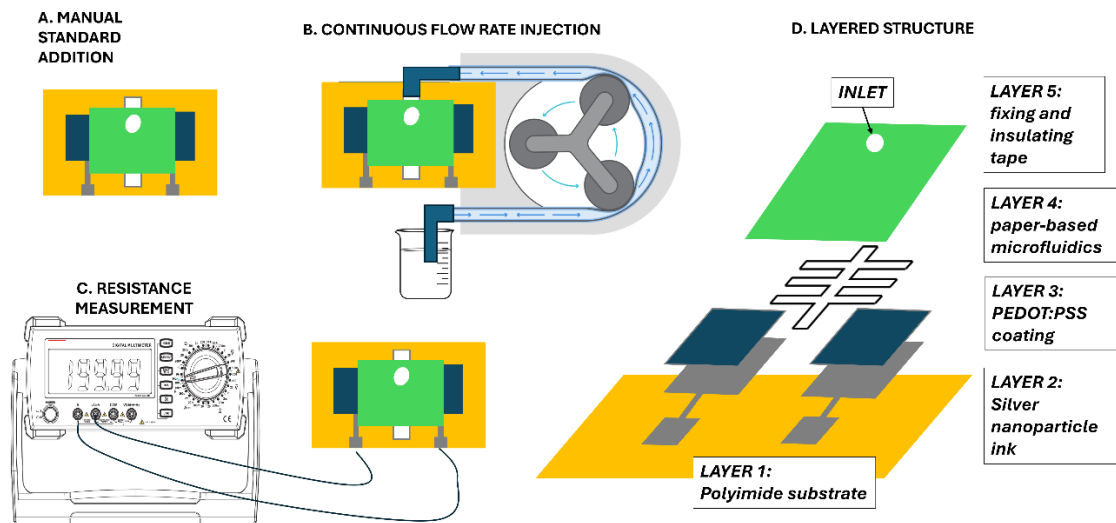


Figure 3: Overview of the fluidic injection methods, electrical characterization, and sensor structure: (A) manual standard addition method for testing with controlled PBS volumes; (B) continuous flow-rate delivery system for testing with controlled PBS rates; (C) resistance measurement setup using a digital multimeter; (D) 3D exploded layout showing the five functional layers of the flexible paper-based microfluidic device

D. In vivo validation of printed electrodes

The in vivo validation was structured into three phases: (i) impedance comparison with commercial electrodes, (ii) controlled laboratory EDA tests (without and with motion artifact interference), and (iii) real-scenario EDA monitoring during work activity.

These experiments were designed to assess both tonic and phasic components of the electrodermal activity (EDA) signal and to verify the suitability of the printed electrodes for wearable use.

To evaluate the applicability of the printed sensors, the contact impedance of the silver/PEDOT:PSS printed electrodes was compared with that of widely used commercial gel electrodes. The purpose was to confirm that the dry printed electrodes provided sufficiently low contact impedance to support proper sensitivity during EDA measurements. Bio-impedance measurements were performed on healthy skin using an Impedance Analyzer (Zurich Instruments) under two-terminal conditions across the 0.5–50,000 Hz frequency range. Impedance values were collected five minutes after electrode placement to allow stabilization of the skin–electrode interface. The participant was seated comfortably with the wrist resting on a table, and four repeated measurements were conducted for both electrode types to assess repeatability.

Subsequently, in vivo measurements were performed in a controlled laboratory environment on two volunteers, seated with their non-dominant wrist resting on a desk. Each participant underwent three recordings:

(1) 60 seconds of baseline followed by a forced expiration to evoke an EDA response.

(2) and (3) three-minute recordings in which the participant was randomly instructed to perform a forced expiration.

Forced expiration was used since it is reported to induce mild sympathetic activation through transient breath-holding and changes in intrathoracic pressure, making it a suitable physiological stimulus for validating EDA response detection [32].

In addition to this acquisition with the subject sitting still, three additional laboratory recordings were conducted to assess sensor performance under controlled motion conditions. During these acquisitions, the participant executed everyday motor tasks while performing a forced-exhalation event. In each trial, both the printed and commercial EDA sensors were simultaneously recorded. More in detail, the first trial began from a standing posture, followed by a cabinet-rearranging activity, a forced exhalation, continued upper-body movement, and subsequent seated rest. The second trial was performed while seated and consisted of a repetitive pick-and-place task on a desk, with a forced exhalation executed during the activity. In the third trial, the participant typed on a computer keyboard, performed a forced exhalation during typing, and then remained at rest without strict control over hand movements. These tasks were selected to introduce controlled yet naturalistic movement artifacts spanning time scales comparable to typical EDA responses, thereby allowing evaluation of the response of the printed sensor against motion-induced interference.

After the laboratory tests, to evaluate the effectiveness of the electrodes for real-world use, preliminary in vivo trials were performed on three operators, each engaged in a distinct industrial activity; without loss of generality, we specifically included resin-coating, kitting, and board rework. For each task, the operator remained still for one minute to record baseline data and then continued their standard working activity. During all in vivo acquisitions, the printed electrodes were positioned on the wrist adjacent to two pre-gelled Ag/AgCl commercial electrodes used as reference. Data were recorded using a commercial multichannel portable device (Plux Biosignal), enabling simultaneous acquisition from three channels: (i) reference Ag/AgCl electrodes for EDA, (ii) the printed electrode pair for EDA, and (iii) the printed electrodes combined with the paper-based microfluidics structure for sweat monitoring. All recordings were stored locally and analyzed offline using a custom MATLAB script.

Final version of paper TIM-26-04647

E. Data processing and quantitative metrics

Regarding the characterization of sweat sensors, the time-dependent resistance data acquired during laboratory tests were converted into conductance values by taking the inverse of each measurement. To improve data quality, a moving-average filter with 1s window was applied to smooth high-frequency fluctuations, and a median filter was used to remove transient artifacts caused by interruptions during successive fluid injections.

To construct the calibration curve, representative conductance values were extracted using a first-derivative thresholding method. Injection events were identified by detecting points on the derivative trace that exceeded a dynamically adjusted threshold, defined as 5 times the standard deviation of the baseline noise recorded prior to the injection. To ensure signal stabilization, the sampling point was not fixed arbitrarily but was set when the absolute value of the first derivative fell below a stability threshold ($< 0.05 \mu\text{S/s}$) for at least 5 consecutive seconds. Preliminary analysis confirmed that this stability criterion was consistently met within 30 seconds post-injection, window that ensured measurements properly recorded only when the signal variance is minimized and the volume-conductance proportionality is stable.

The resulting calibration curve was then used to convert in vivo conductance changes (ΔG) into estimated sweat volumes (ΔV). For each sensor, the slope (expressed in $\mu\text{S}/\mu\text{L}$) was computed via linear regression. During in vivo sessions, the equivalent sweat volume was calculated as:

$$\Delta V = \frac{\Delta G}{S_{\text{cal}}} \quad [1]$$

Raw EDA signals from in vivo recordings were processed using LedaLab (v3.4.9) within a MATLAB environment [33]. The signals were first low-pass filtered using a 4th-order Butterworth filter (5 Hz cutoff) and smoothed with a 20-sample moving-average filter. To separate the tonic (Skin Conductance Level, SCL) and phasic (Skin Conductance Response, SCR) components, we employed Continuous Decomposition Analysis (CDA) with a standard deconvolution kernel. The CDA optimization was executed with two initial iterations to estimate the tau parameters, modeling the EDA signal as the convolution of sudomotor nerve impulses with a physiological impulse response. This method provides continuous estimates of sudomotor nerve activity (SMNA) and enables accurate separation of overlapping SCRs. For SCR detection, an adaptive threshold (T_{SCR}) was defined based on the first minute of resting baseline, calculated as:

$$T_{\text{SCR}} = \mu_{\text{baseline}} + 3\sigma_{\text{baseline}} \quad [2]$$

where μ_{baseline} and σ_{baseline} represent the mean and standard deviation of the baseline signal, respectively. This approach ensures balanced sensitivity in detecting SCR for wearable sensors in motion-intensive scenarios.

Quantitative metrics were employed to enable comparison between printed and commercial EDA electrodes. First, morphological and trend similarity were assessed via the Pearson Correlation Coefficient (r), calculated independently for the decomposed tonic (SCL) and phasic (SCR) components. Second, temporal precision and event matching were quantified by pairing SCR onsets (identified via Eq. [2]) within a 1 s tolerance window to calculate the Mean Onset Latency (Δ_{onset}).

Third, amplitude and sensitivity analysis was performed using a Mean Scaling Factor (ratio of printed to commercial SCR peaks) and Linear Regression to verify consistency across response magnitudes. Additionally, to evaluate the system's accuracy in capturing overall sympathetic response rates, the difference in response frequency ($\Delta\text{SCR}/\text{min}$) was computed as follows (Eq.3):

$$\Delta\text{SCR}/\text{min} = \left| \frac{N_{\text{printed}}}{T_{\text{min}}} + \frac{N_{\text{comercial}}}{T_{\text{min}}} \right| \quad [3]$$

where N_{printed} and $N_{\text{comercial}}$ represent the total number of valid SCR peaks detected above the adaptive threshold (Eq. [2]) for the printed and commercial sensors respectively, and T_{min} is the total duration of the in vivo recording session in minutes.

Finally, systematic offset and error in the tonic baseline were defined through the Mean Absolute Error (MAE, representing the average absolute difference in μS between signals) and Mean Percentage Error (MPE, quantifying the average relative discrepancy), accounting for the impact of the dry contact impedance.

III. RESULTS AND DISCUSSIONS

A. Laboratory characterization of printed electrodes

1) Paper grade and substrates comparisons

To extend the linear detection range of the electrodes, several Whatman paper grades were evaluated to investigate the relationship between paper morphology (thickness and pore size) and sensor performance in terms of sensitivity and dynamic range, considering 3 sensors for each conditions. As shown in Figure 4 and Table 1, an increase in paper thickness correlated with enhanced sensitivity. In contrast, larger pore diameters negatively affected calibration-curve sharpness and sensitivity.

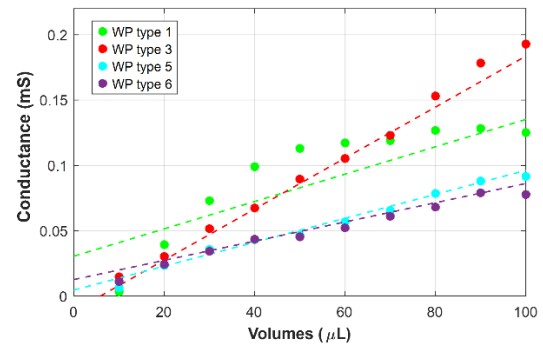


Figure 4. Comparison of obtained results with printed electrodes for using various type of paper grades. The displayed data represents the average value from triplicated measurements.

TABLE 1: RESULTS IN TERMS OF SENSITIVITY OF THE TESTS PERFORMED WITH DIFFERENT PAPER GRADES

	Thickness (μm)	Porosity (μm)	Sensitivity ($\mu\text{S}/\mu\text{L}$)
WP Grade 1	180	11	1.05
WP Grade 3	390	6	1.95
WP Grade 5	200	2.5	0.92
WP Grade 6	180	3	0.73

Final version of paper TIM-26-04647

This behavior can be explained by the differences in fluid retention exposed by each paper design. In fact, papers with smaller pores retain liquid for longer periods, maintaining a stable and uniform contact interface between the wetted paper and the electrodes. In contrast, highly porous papers suffer from rapid fluid transport and evaporation, leading to reduced contact efficiency and lower signal stability.

Based on these results, Whatman Grades 3 and 5 were selected for further evaluation across another distinct batch of five sensors fabricated under identical conditions. As reported in Figure 5.A and Table 2, Grade 3 exhibited higher average sensitivity than Grade 5. Although Grade 3 has a larger pore size, its greater thickness improves liquid-holding capacity, producing steeper calibration slopes. This confirms that paper thickness (i.e., fluid capacity) plays a more dominant role than pore size in determining measurement sensitivity. Therefore, Grade 3 was selected for subsequent experiments.

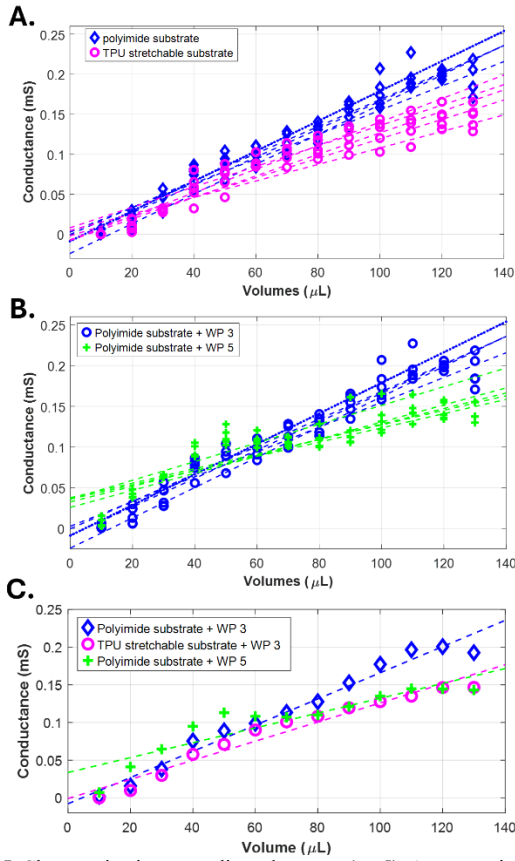


Figure 5. Characterization on replicated sensors ($n=5$): A. comparison between different paper grades (3 and 5) on Kapton substrate; B. comparison between different substrates, Kapton and Intexar; C. comparison among the average values of the three conditions: polyimide+WP3, TPU+WP3 and polyimide+WP5

TABLE 2: RESULTS IN TERMS OF SLOPE OF THE REPETITIVE TESTS PERFORMED WITH MULTIPLE SENSORS ON KAPTON SUBSTRATES WITH PAPER GRADES 3 AND 5, AND ON TPU STRETCHABLE SUBSTRATE FOR PAPER GRADE 3.

	S1	S2	S3	S4	S5	Average ($\mu\text{S}/\mu\text{L}$)	RSD (%)
Polyimide WP 3	1.74	1.88	1.86	1.52	1.69	1.74	8.3
Polyimide WP 5	0.96	0.87	0.89	1.05	1.15	0.98	11.6
Stretch TPU WP 3	1.53	1.66	1.62	1.16	1.45	1.48	13.4

Repeated measurements on five devices confirmed the superior performance of Grade 3 in terms of both sensitivity and linearity range. The average sensitivity and relative standard deviation (RSD) for each condition are summarized in Table 1, demonstrating consistent behavior across repetitions.

To evaluate integration within stretchable wearable platforms, printed electrodes were also fabricated on a thermoplastic polyurethane (TPU) substrate (Intexar™, Micromax) and compared with those printed on polyimide. As shown in Figure 5B, TPU-based sensors demonstrated slightly lower sensitivity; however, the values remained within an acceptable range for reliable operation. Additionally, RSD values were comparable between TPU and polyimide devices, confirming that the fabrication process and selected materials are suitable for flexible and wearable configurations. While the present design is optimized for high sensitivity in the 0–50 μL range, the absorption limit of the Whatman paper substrate represents a constraint for long-term monitoring. To bypass this limitation, future developments will investigate the inclusion of an integrated waste reservoir to prevent fluid accumulation at the electrodes. This architectural enhancement will allow for continuous sweat volume tracking beyond the current saturation threshold, ensuring performance stability across a broader spectrum of physiological conditions.

2) Characterization under controlled flow rate

Measurements performed under different controlled flow rates ($\mu\text{L}/\text{min}$) demonstrated that the electrodes combined with the paper-based microfluidic network can discriminate between varying flow conditions (Figure 6). The slope of the initial rising region exhibited a linear correlation with flow rate, with an average sensitivity of approximately $1.4 (\mu\text{S}/\text{min})/(\mu\text{L}/\text{min})$. These results align with previous findings [34], which report a monotonic linear increase in signal with sensitivity proportional to applied flow rate until plateauing, potentially indicating that the substrate is approaching its fluid-handling capacity. Upon reaching the limit of linear range, higher flow rates produced a pronounced peak corresponding to the maximum absorbable volume, followed by a decrease towards a plateau. This behavior was likely caused by a mismatch between the volume delivered by the pump and the absorption capacity of the paper matrix. Once the paper became locally saturated, excess fluid temporarily accumulated before dispersing, reducing the effective conductance interface. In contrast, at lower flow rates, after a gradual increase, the signal transitioned directly into a plateau without any overshoot, suggesting that the system efficiently managed lower rates of fluid delivery without saturation events.

The in vitro results demonstrate that the proposed patch can reliably discriminate between flow rates in the 10–30 $\mu\text{L}/\text{min}$ range, which corresponds to the low-to-moderate sweating levels typically observed during cognitive stress or light physical activity in industrial settings. By establishing this quantitative relationship between fluid volume and electrical conductance in a controlled environment, we provide the necessary validation for interpreting subsequent in vivo signals. Specifically, the linear conductance-to-flow-rate correlation observed here serves as the physiological proxy required to properly estimate physical sweat volume from the tonic shifts in skin conductance recorded during real-world tasks.

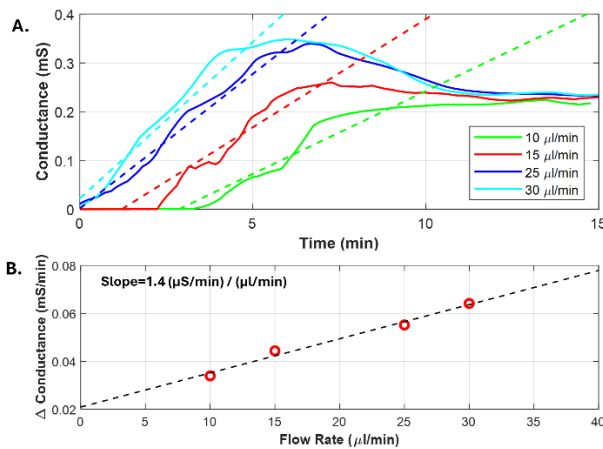


Figure 6. Calibration of the paper-based sweat-volume sensor under controlled flow rates (10–30 $\mu\text{L}/\text{min}$). A. Electrical conductance response over time under different imposed flow rates, illustrating the transient behavior, capillary overshoot, and subsequent stabilization/saturation at higher fluidic loads. B. Linear relationship between conductance slope and flow rate.

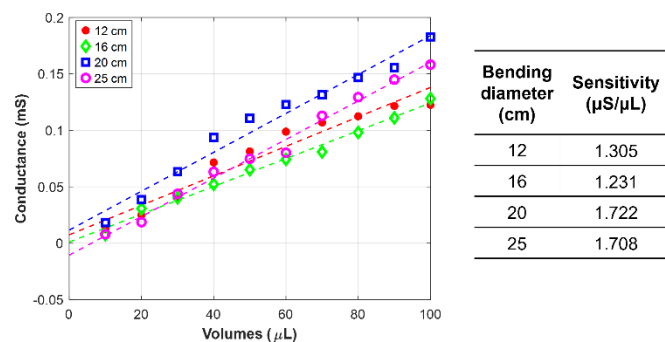


Figure 7. Calibration curve obtained from fully printed electrodes under bending conditions

3) Bending tests

Since the sensor is intended for wearable applications, its performance under bending was assessed by mounting devices on cylindrical supports with different diameters. As shown in Figure 7, the highest bending condition (12 cm diameter) resulted in the lowest sensitivity, with a calibration slope of $1.057 \mu\text{S}/\mu\text{L}$. This reduction is likely to be mainly attributed to the non-uniform pressure distribution across the device layers during curvature, which affects the interface contact resistance between the inkjet-printed electrode and the paper millifluidic network. Additionally, bending induces a localized compression of the paper pores, slightly altering the fluid distribution and the effective electrochemical surface area in contact with the electrodes. As the cylinder diameter increased and curvature decreased, sensitivity progressively approached the values obtained under flat conditions. Although bending introduced a modest reduction in sensitivity, sensor functionality and calibration integrity were preserved. To mitigate these effects, future design optimizations could include the use of serpentine-shaped electrodes to better accommodate mechanical strain or the application of a compliant encapsulation layer or adhesive to ensure a more stable and constant contact pressure throughout the sensing area.

B. In vivo validation of printed electrodes

1) Bioimpedance

As can be observed from the obtained results (Fig. 8), the Nyquist plot displays a typical semicircular shape, where the diameter of the semicircle reflects the contribution of the dominant electrode-skin interfacial resistive component and the stratum corneum resistance under an equivalent-circuit interpretation.

As expected, a smaller diameter of the semicircle for gel electrodes can be observed, indicating lower impedance and therefore better conductivity, overall. This reduced impedance could be mainly due to the presence of gel, which improves skin-electrode contact by filling microscopic gaps at the interface, thus enhancing ionic conduction and reducing electrode-skin resistance [35]. The impedance magnitude at low frequency was also quantified to compare the effective skin-electrode coupling. At 1 Hz, the average impedance for the commercial Ag/AgCl electrodes was $3000 \text{ k}\Omega$, while the printed Ag/PEDOT:PSS electrodes exhibited $5500 \text{ k}\Omega$ ($n = 4$ repetitions per condition). The relative difference between the two configurations remained within the same order of magnitude; the maximum deviation observed at 1 Hz ($\sim 83\%$) rapidly decreases at higher frequencies and does not compromise the signal integrity for EDA applications in the 0.05–5 Hz band.

On the other hand, printed electrodes - being dry - exhibit a higher contact impedance and an average RSD of $\sim 10\%$, compared to $\sim 1\%$ for commercial gel electrodes. This behavior is consistent with the intrinsically greater variability of dry interfaces and with the increased likelihood of relative movement between skin and electrodes during acquisition [36]. Nevertheless, the obtained values remain acceptable for wearable EDA acquisition. To mitigate issues related to higher contact impedance and variability, several strategies were adopted for in vivo testing. First, mechanical stability at the skin interface was improved using wearable straps that secure the printed electrodes and minimize motion artifacts. Second, complete insulation of all conductive tracks was implemented to avoid accidental contact with external conductive surfaces, which could otherwise alter impedance measurements and degrade signal quality.

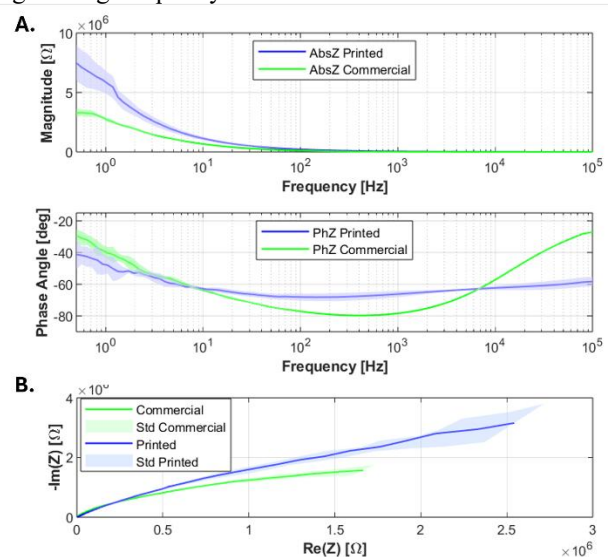


Figure 8. Bioimpedance measurement using commercial gel electrodes and fully printed dry electrodes: A. Magnitude and Phase angle versus frequency and B. Nyquist plot

2) Laboratory tests

Laboratory-based experiments reported comparable EDA trends between commercial pre-gelled electrodes and the newly developed printed ones (Fig. 9 and 10). These results align with findings highlighting that PEDOT:PSS coated silver electrodes can capture dynamic features similarly to traditional Ag/AgCl wet electrodes [37]. This agreement was consistent across all repetitions and subjects, despite the expected inter-subject variability in EDA amplitude and reactivity [35], which was observed for both electrode types.

Regarding signal magnitude, the overall estimated average EDA conductivity for commercial electrodes was higher, as highlighted by an average tonic MPE of 57% and phasic SCR scaling factor of 0.66. This behavior, typical of dry electrodes [38], is attributed to the enhanced ionic coupling of gel interfaces, which results in lower contact impedance compared to dry interfaces [39]. The offset in tonic levels observed is frequently reported for thin-film inks due to higher contact resistance compared to electrolytic gels [40]. This observation is consistent with bioimpedance measurements, where printed electrodes exhibited higher resistance (relative difference at 1 Hz ~83%), particularly at low frequencies. Given that EDA signals are measured under quasi-DC conditions, this higher resistance explains the lower tonic conductance values obtained with the printed electrodes (average tonic difference with a MAE of $2.56 \pm 2 \mu\text{S}$).

Despite these amplitude differences, no marked discrepancies were observed in the number or temporal occurrence of phasic responses. The printed electrodes detected sympathetic-related peaks with a timing accuracy computed as Mean Onset Latency of 73 ± 100 ms. This is consistent with studies on flexible dry interfaces, which indicate that high correlation with gold-standard Ag/AgCl sensors can be maintained for phasic detection despite variations in contact impedance [38]. Specifically, stress-induced SCR peaks were detected by both systems with a comparable number of events N ($N_{\text{printed}} = 11 \pm 3$ vs $N_{\text{comm}} = 10 \pm 4$ for subject 1 and $N_{\text{printed}} = 23 \pm 3$ vs $N_{\text{comm}} = 22 \pm 3$ for subject 2). As shown in Fig. 10, In Fig. 10, the

values obtained with printed electrodes matched those from commercial electrodes with a discrepancy lower than 10%. This performance aligns with the State-of-the-Art in printed wearable electronics, where silver-based flexible sensors have been shown capable of replicating the signal morphology of Ag/AgCl electrodes during cognitive stress [41]. Although cumulative SCR amplitudes were lower for printed electrodes (SCR scaling factors of 0.71 and 0.61 for subject 1 and 2 respectively) the scaling of the signal did not compromise the ability to track physiological trends [42]. As shown in Fig. 9, signals recorded with printed electrodes followed the trends of commercial electrodes with the following correlation coefficients: $r_{\text{tonic}} = 0.82 \pm 0.14$ and $r_{\text{phasic}} = 0.76 \pm 0.05$ for subject 1 and $r_{\text{tonic}} = 0.83 \pm 0.07$ and $r_{\text{phasic}} = 0.66 \pm 0.19$ for subject 2. Furthermore, evaluating the peak frequency across the monitoring sessions yielded a mean response difference ($\Delta\text{SCR}/\text{min}$) of less than 3 responses per minute (2.1 ± 0.8 SCR/min) between the printed dry patch and the commercial gel-based electrodes, suggesting a comparable performance in tracking discrete sudomotor events, despite differences in contact impedance. While the absolute magnitude of SCL and SCR is influenced by interface properties, the detection of phasic events is suggested as a potentially robust metric independent of the specific measurement method [43]. Within each subject, the relative amplitude of responses remained stable across the three acquisitions (RSD of both tonic MPE and SCR scaling factor $< 15\%$), confirming a promising starting point in term of repeatability of the measurements under controlled conditions [44]. The stability of the signal across sessions suggests that the printed ink-polyimide adhesion is robust for short-term acquisition, though long-term stability may require further optimization to prevent signal degradation [45]. Finally, substantial inter-subject variability was observed mainly in baseline conductance levels and in evoked responses [35], [46].

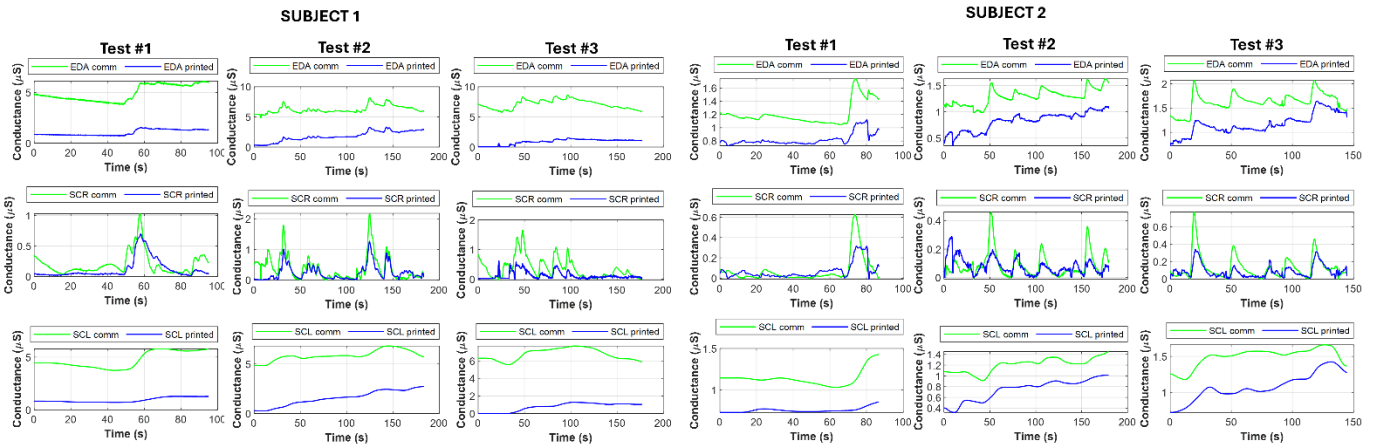


Figure 9. Comparison between commercial pre-gelled electrodes (green) and printed dry electrodes (blue) during repeated controlled laboratory tasks on subject 1 and 2. Top row: raw EDA signals; middle row: phasic SCR obtained via decomposition; bottom row: tonic Skin Conductance Level (SCL). Each column corresponds to a separate acquisition session as named by titles test 1, 2 and 3.

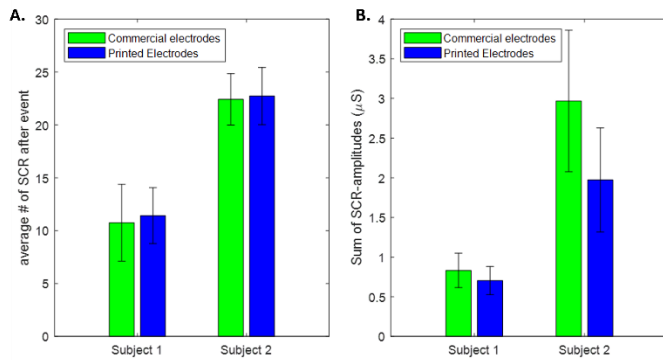


Figure 10. Comparison of Skin Conductance Response (SCR) features obtained from commercial pre-gelled electrodes (green) and printed dry electrodes (blue) for two subjects. A. average number of SCRs detected following each stimulation event. B. cumulative SCR amplitude (μS) across events.

This is largely due to the inherent subject-dependent nature of EDA signals (marked differences in number of events, average EDA, and MPE values (80% vs. 34%)) likely exacerbated by differences in sensor wearability. These results highlight that while printed sensors are promising, future refinements must prioritize ergonomic consistency and subject-specific calibration to minimize motion artifacts and ensure reliable data across diverse users. Figure 11 summarizes the results of laboratory acquisitions where forced expiration was intentionally superimposed on motion artifacts. Across three trials, subjects performed the same respiratory task while engaged in typical office activities, such as rearranging objects, pick-and-place desk work, and typing. In all conditions, movement artifacts appeared comparable in amplitude and temporal structure across both sensor types, suggesting that the printed electrodes exhibited a vulnerability to motion-induced fluctuations comparable to that of commercial ones. This observation aligns with previous literature suggesting that EDA signals, even when motion-affected, can retain noticeable within-subject correlations for both tonic and phasic components across different body sites [47].

To quantitatively assess these impacts, a comparative analysis was performed (detailed in Supplementary Data, Figs. 1-3 and Table 1). The printed electrodes suggested satisfactory morphological and temporal consistency with commercial standards, yielding phasic EDA correlations with $r > 0.6$ (r of 0.95, 0.67, and 0.68 for tasks 1, 2, and 3, respectively) and mean onset latency below 1 s (Δ of 383, 450 and 550 ms for task 1,2,3 respectively.). It was then quantified a signal attenuation of the dry interface which resulted in an average SCR scaling factor (ratio between the printed and commercial peak amplitudes) of 0.72 ± 0.20 and in a Tonic Mean Percentage Error (MPE, %), (relative discrepancy between the tonic baselines) of $52 \pm 20\%$. A simplified phasic SNR calculation was also performed by treating all signal components above 1 Hz as noise from artifacts, obtaining similar average phasic SNR ($\Delta < 2$ dB). While these preliminary results potentially support the technical feasibility of the presented solution for appreciating EDA trends during light motion, we acknowledge that a rigorous and

reliable conclusion about distinguishing physiological SCRs from

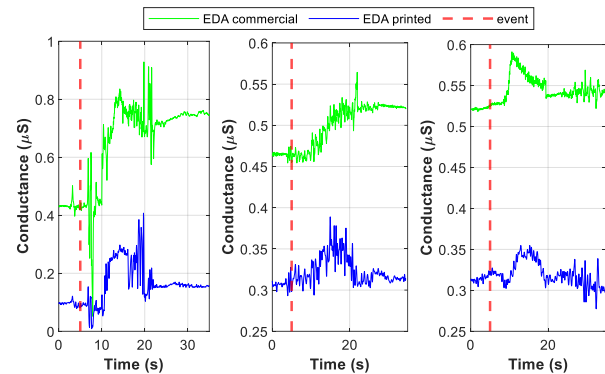


Figure 11. EDA responses during three repeated trials in which forced-exhalation events (red dashed line) were performed not at rest, but during controlled movement artifacts. Specific artifacts were A. cabinet-rearranging activity, B. seated pick-and-place task at a desk, C. typing at a computer + standard hand movements. Each panel shows both the commercial EDA signal (green) and the printed-sensor EDA signal (blue).

motion-induced fluctuations remains challenging due to overlapping times and frequency bands [47], [48], and could be stated only after future in depth analysis employing more sophisticated techniques. Previous research has emphasized that since EDA responses evolve over slow physiological timescales (≈ 10 – 20 s), motion events of similar duration present a considerable confound, a phenomenon observed in both modalities during our tests [49], [50]. Despite the preliminary results and analysis reported that EDA changes remain detectable even when superimposed on motion, current limitations emerge under prolonged or non-stationary movement. To mitigate this in future designs, electrode geometry could be optimized for better skin conformity, and more compliant, stretchable substrates could replace polyimide to enhance mechanical coupling [51]. Furthermore, from an electronic standpoint, integrating high-impedance, unity-gain buffer amplifiers directly at the sensing site could minimize cable-related interference and signal attenuation, representing the current frontier in high-fidelity ambulatory monitoring [52]. Developing such advanced hardware, alongside more sophisticated signal processing methodologies, is the focus of our ongoing research to further validate sensor performance in dynamic conditions.

3) Real case scenario test

A real-world validation was conducted on three distinct operators, each performing one of three continuous manufacturing tasks: 1) resin coating of electronic components, 2) board rework activities alternating between soldering and microscope inspection, and 3) component kitting for automated board fabrication (Fig.12.A). These tasks represent an increasing gradient of physical activity: a light-activity industrial workflow with natural arm and wrist movements (Task 1); alternating static and dynamic conditions (Task 2); and an overall dynamic condition involving walking and full-body movements (Task 3). This pilot study serves as a baseline for future operational testing, providing realistic motion

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challenges while retaining physiological relevance for stress and workload monitoring.

The integrated printed patch supported multimodal biosensing, combining electrodermal activity (EDA) measurement with millifluidic sweat-volume monitoring (Fig. 12 B-C). As summarized in Table 3 (and extensively reported in Supplementary Fig. S3), the printed EDA electrodes exhibited comparable trends with commercial gel electrodes, suggesting the capability to track sympathetic dynamics trends even under uncontrolled conditions.

The higher contact impedance of the printed electrodes primarily resulted in a reduced absolute magnitude of the tonic signal compared to gel electrodes (tonic MPE of 52%, 62% and 45% for task 1, 2, 3 respectively). However, this did not result in missed physiological events: there was no marked discrepancy in the timing or number of Skin Conductance Responses (SCRs), with an average Δ_{onset} of 108 ms and consistent phasic correlation coefficient of 0.75 ± 0.10 . This temporal consistency during motion tasks (Fig. 11) suggests that the mechanical stability of the patch compensates for the absence of conductive gel, preserving the signal morphology and maintaining a consistent correlation for both tonic (SCL) and phasic (SCR) components (average $r_{\text{tonic}} = 0.64 \pm 0.14$; average $r_{\text{phasic}} = 0.75 \pm 0.10$).

TABLE 3: QUANTITATIVE PARAMETERS FOR ANALYSIS OF EDA SIGNALS ACQUIRED FROM COMMERCIAL AND PRINTED ELECTRODES DURING REAL CASE SCENARIO TASKS

Task	R_{TONIC}	R_{PHASIC}	Δ_{onset} (MS)	TONIC MPE (%)	SCR SCALING FACTOR
Resin Coating	0.523	0.668	10.75	52.07%	0.456
Rework	0.793	0.868	44.04	62.39%	0.491
Kitting	0.613	0.702	270	45.03%	0.662

Regarding the paper-based millifluidic sweat sensor, the response obtained during the three tasks suggested a correlation between physical intensity and sweat dynamics. In Task 1, involving lighter activity, the device remained inactive for the initial ~ 1000 s, eventually reaching a conductance of $\sim 2.5 \mu\text{S}$ ($\sim 1.47 \mu\text{L}$, based on the sensitivity of $1.74 \mu\text{S}/\mu\text{L}$ provided in section III.A.1 from laboratory calibration). This latency period aligns with the physiological induction time required for sweat accumulation in paper-based wearable microfluidic devices under low-intensity activity (around 10 min) [53]. In contrast, the more dynamic Tasks 2 and 3 showed reduced latency periods (~ 200 s and ~ 50 s, respectively), likely due to a faster onset of sweating due to increased metabolic demand. Quantitatively, Task 2 reached $\sim 1.8 \mu\text{S}$ ($\sim 1.06 \mu\text{L}$) with a steeper rise after induction, suggesting a higher sweat rate during board rework compared to Task 1. Task 3 exhibited the highest accumulation, reaching $\sim 4.5 \mu\text{S}$ ($\sim 2.65 \mu\text{L}$) with a nearly constant slope that points to a sustained sweat rate during continuous movement. While the conductance slope offers a valuable proxy for the sweat rate, these trends must be interpreted with caution due to potential local micro-

environmental fluctuations. Overall, these results represent a preliminary feasibility test proposing the sensor as a useful cumulative detector, suggesting the possibility to estimate quantitative thermoregulatory data that complements EDA for potential future workload monitoring.

Movement can introduce drift in sweat rate measurements by briefly altering the contact between the skin, inlet, and paper channel. In our design, the millifluidic network passively stabilizes the interface by wicking sweat away from the skin, making the signal less sensitive to short mechanical disturbances. The applied smoothing and median filters further suppress rapid fluctuations, while the derivative-based event detection naturally rejects slow drift. Future wearable versions will incorporate additional strategies—such as soft mechanical decoupling layers, reference channels, and adaptive drift compensation—to further reduce motion artifacts during dynamic tasks. Moreover, porous materials alternative to paper are being studied for future replicas to enhance sensitivity in resting or near-resting conditions [54]. This technical approach is crucial because, as observed during the more dynamic Tasks 2 and 3, spontaneous SCR peaks can sometimes be superimposed on movement-induced fluctuations of comparable amplitude. This underscores a critical challenge in the state of the art: the spectral and temporal overlap between autonomic signals and physical movement in ambulatory settings [48]. To further minimize this sensitivity, future developments will focus on refining electrode geometry and substrate properties—such as exploring more conformable and stretchable materials—to improve mechanical coupling and reduce shear forces [51]. Overall, while a broader validation on a larger cohort of operators is strictly necessary to confirm generalizability, these preliminary results from three subjects demonstrate a promising field feasibility for operation under real-use conditions, representing a relevant starting point for advancing wearable systems toward practical multimodal physiological monitoring in occupational environments.

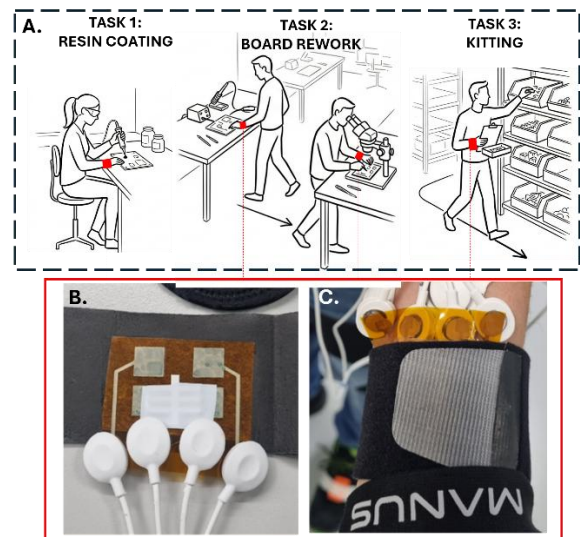


Figure 12. Overview of the tasks (A.) monitored through the wearable bracelet and photographs of the wearable patch prototype, including the printed electrodes interfaced with standard snap connectors (B.) and the patch worn on the forearm under a sport sleeve during testing (C.).

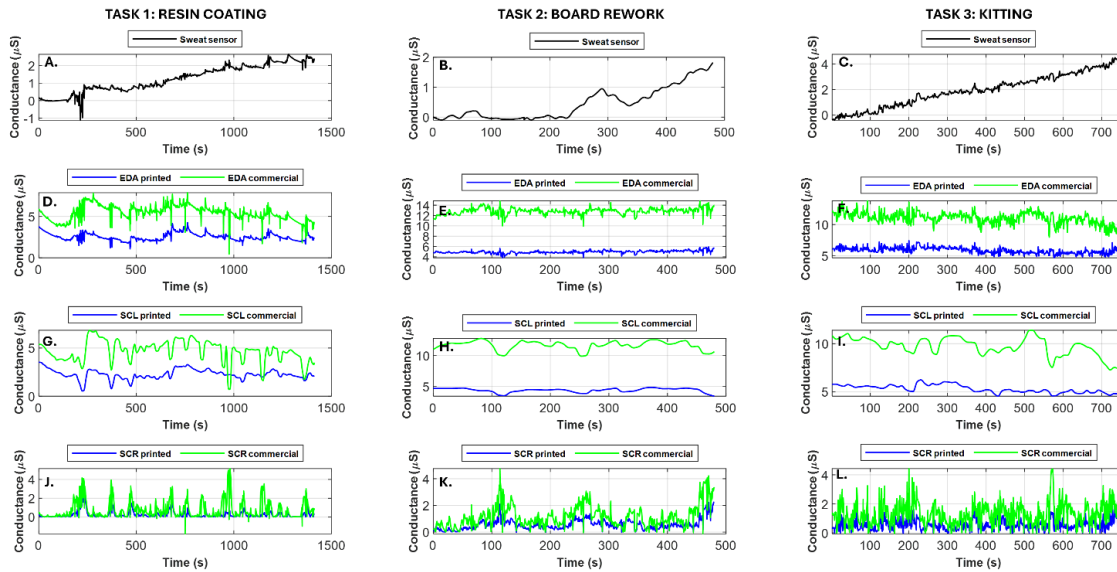


Figure 13. Results obtained with the integrated multisensor patch combining printed EDA electrodes and a paper-based millifluidic sweat-volume sensor during the three tasks monitored in a real-world use scenario. A,B,C. Conductance-based electrical proxy tracking sweat accumulation dynamics during the real-scenario protocol. D-L Simultaneous EDA measurements from the printed sensor (blue) and commercial electrodes (green), including the overall EDA waveform (D,E,F), its tonic component (Skin Conductance Level, SCL) (G,H,I), and phasic component, SCR (J,K,L).

IV. CONCLUSION AND FUTURE PERSPECTIVES

This work presented a preliminary feasibility study of a fully printed multimodal patch for the simultaneous monitoring of electrodermal activity (EDA), sweat accumulation, and sweat rate. By leveraging inkjet-printed electrodes integrated with paper-based millifluidics, the device demonstrated the capability to capture both sympathetic nervous system triggers and local sweat accumulation dynamics.

Regarding EDA monitoring, the higher contact impedance of the dry printed interface primarily affects the absolute SCL, introducing a systematic offset (average MPE of 55%). However, this mismatch appears not to hinder the extraction of psychophysiological features: the sensor maintains acceptable data fidelity for preliminary testing in the phasic component, with average r_{phasic} ranging from 0.71 to 0.77 in the different acquisition contexts and a temporal accuracy characterized by Mean Onset Latency below 600 ms. These results suggest that the printed electrodes are highly effective for applications prioritizing sympathetic arousal and peak-based features over absolute tonic quantification. Concerning sweat-sensing, *in vitro* tests confirmed a clear discrimination of PBS volumes (0–120 μL) and flow rates (10–30 $\mu\text{L}/\text{min}$), translating to linear changes in conductance magnitude (μS) and slope ($\mu\text{S}/\text{min}$). In real-case scenarios, the patch successfully captured sweat accumulation correlating with physical effort, confirming the complementary value of integrating hydration-related data with autonomic monitoring. The cleanroom-free, low-cost printing approach offers high geometric flexibility for seamless integration into garments, wrist supports, or professional PPE. However, practical long-term use requires addressing several reliability and system-level factors. First, to mitigate the effects of repeated bending, prolonged sweat exposure, and electrolyte buildup on the paper's capillary behavior, a multi-layered packaging approach will be implemented using wet-strength papers and encapsulation with thin, breathable hydrophobic

films (e.g., TPU) featuring a dedicated opening for the millifluidic inlet [55]. Second, while commercial laboratory equipment was used here to isolate the intrinsic performance of the sensing layer, the architecture is fully compatible with low-power wearable electronics. Future hardware implementations will embed a high-input-impedance amplifier ($\geq 10\text{ M}\Omega$) to preserve low-frequency EDA content, a low-pass filter ($\sim 5\text{ Hz}$), and a stable DC readout for sweat monitoring. To ensure signal integrity in high-workload settings, these hardware improvements will be complemented by advanced digital signal processing. Although a formal correction algorithm was not developed in this preliminary study - which was rather focused on sensor fabrication and preliminary testing - future work will address the design of mathematical adaptive filters (e.g., Recursive Least Squares) and machine learning classifiers (e.g., 1D U-Net architectures) to dynamically mitigate motion artifacts and automatically isolate sweat artifacts using the concurrent baseline streams provided by this multi-sensing configuration [56].

Finally, the small participant pool limits the generalizability of these findings. Future larger-scale clinical tests against reference devices (e.g., Macroduct) are necessary to account for inter-subject variability. Furthermore, while the device reliably identifies perspiration trends, achieving absolute volumetric accuracy remains challenging; future studies employing *in-situ* optical imaging or integrated pressure sensors will be essential to decouple skin-interface effects from physiological sweat dynamics. In conclusion, while further integration of advanced packaging, artifact-aware hardware, and dedicated software layers remains a focus for future investigations, the preliminary results obtained in this study offer a promising starting point and a valuable building block toward next-generation wearable systems for simultaneous electrodermal activity and local sweat dynamics monitoring, which could potentially support stress or workload assessment when combined with multi-sensor platforms in occupational environments.

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